NAME: ____ CLASS: _________ INDEX: __________

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

Answers

BIOLOGY
8875/01

Paper 1 Multiple Choice

2nd September 2014
1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the Answer Sheet in the spaces provided unless this has been done for you.

There are **30 MCQ** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **2B pencil** on the separate Answer Sheet.

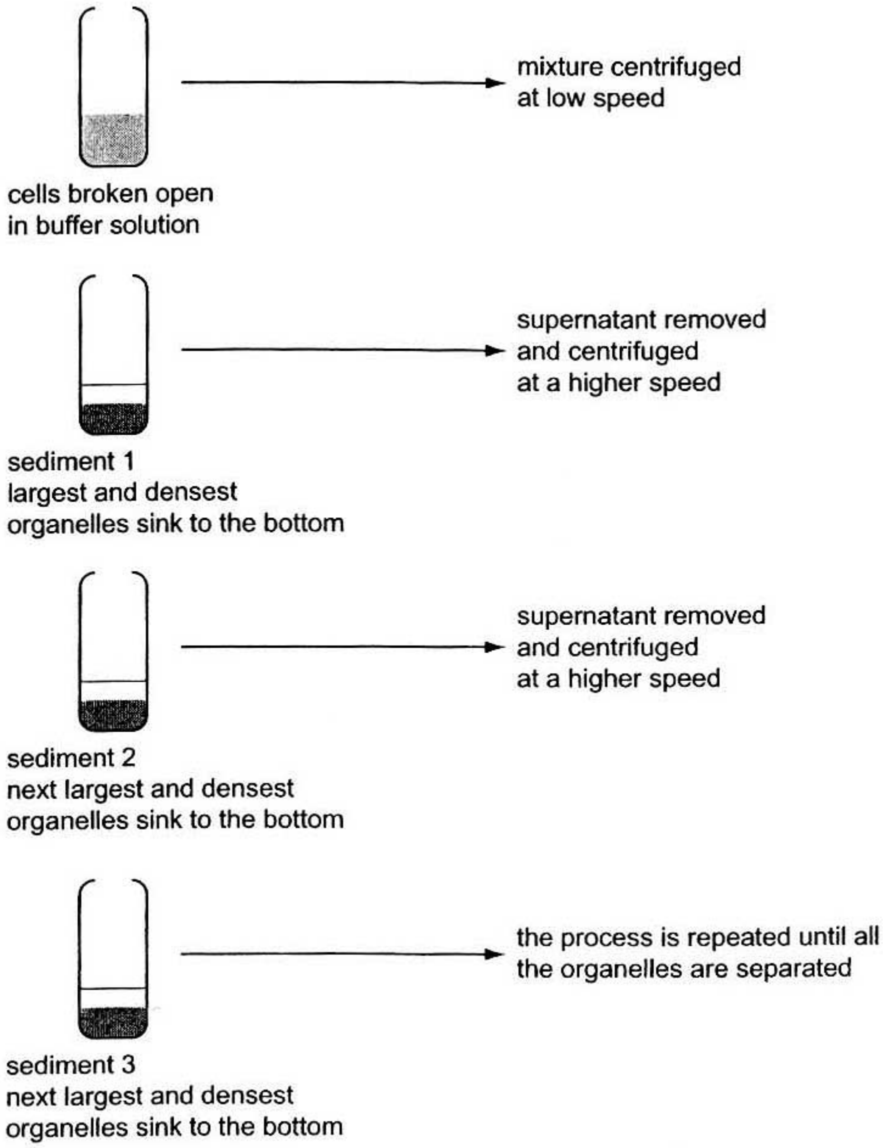
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

1	D	2	B	3	B	4	D	5	B
6	B	7	B	8	C	9	B	10	C
11	C	12	B	13	D	14	C	15	C
16	C	17	A	18	C	19	C	20	C
21	B	22	A	23	D	24	C	25	D
26	B	27	C	28	D	29	C	30	A

1. Fractionation is a process used to separate cell components according to their size and density.

The diagram below shows the main steps in fractionation of plant cells.



DCPIP and buffer solution were added to each sediment and the mixtures left in the light for fifteen minutes. Sediment 2 caused the oxidised DCPIP to be reduced.

Which organelles are present in sediment 2?

A ribosomes

B mitochondria

C nuclei

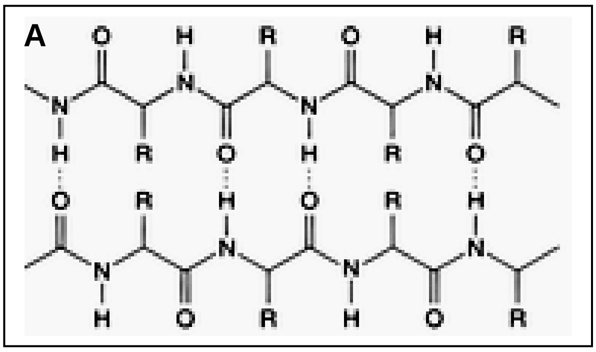
D chloroplasts

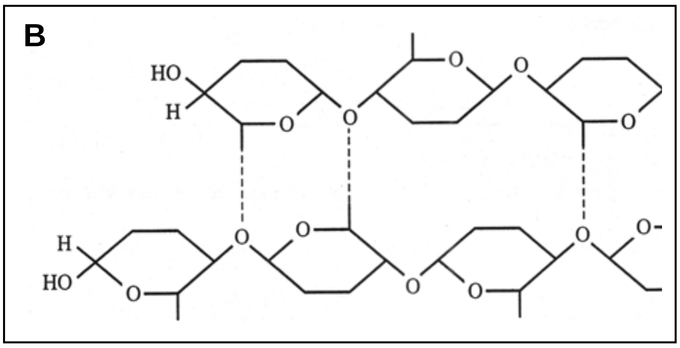
Ans: D

[L2] (*MJC 09/P1/01*)

SC: Process determined estimate of organelle size & density, Sediment 2 cause reduction of DCPIP.
ORE: Only chloroplast fulfil the context of qn.

2. The following shows two molecules that perform structural functions.





Which of the following statements does not describe a similarity between molecules A and B?

A Hydrogen bonds form cross-linkages between neighbouring chains.

B Monomers are joined together via peptide bonds.

C Monomers can be extracted by hydrolysing the bonds within each chain.

D Covalent bonds are present in both molecules.

Ans: B

[L2] (*11TJC/P1/7*)

SC: A, C and D are shown in the two molecules,
ORE: Hence Option B is not the similarity

3. The **R** group of the amino acid *serine* is $-CH_2-OH$. The **R** group of the amino acid *alanine* is $-CH_3$. Where would you expect to find these amino acids in a globular protein in aqueous solution?

A Serine would be in the interior, and alanine would be on the exterior of the globular protein.

B Alanine would be in the interior, and serine would be on the exterior of the globular protein.

C Both serine and alanine would be in the interior and on the exterior of the globular protein

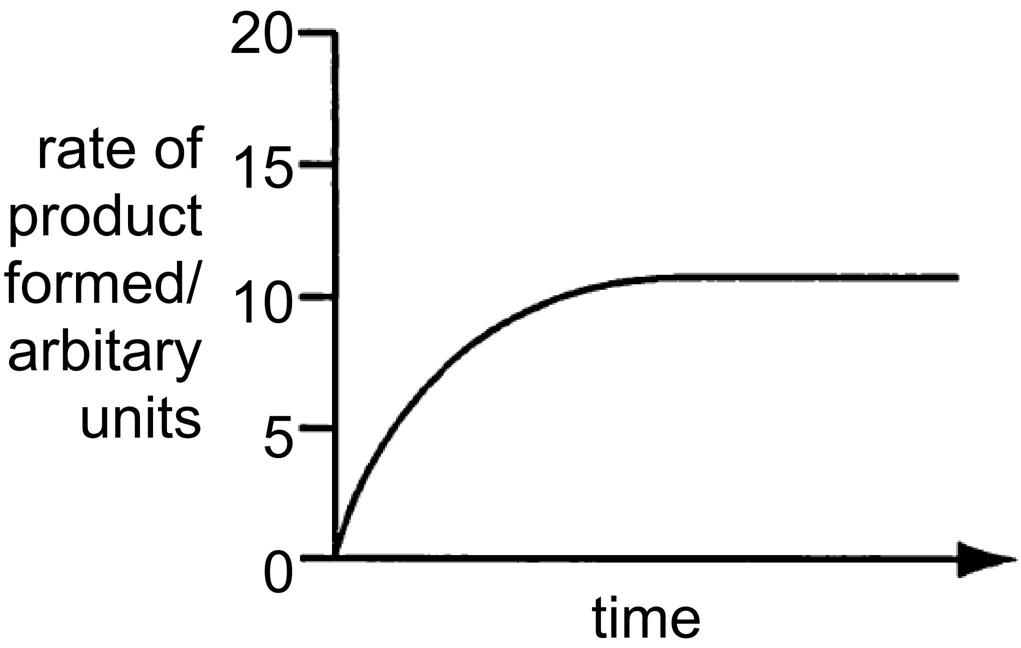
D Both serine and alanine would be in the interior of the globular protein.

Ans: B

[L2] (*11IJC/P1/3*)

SC: Serine has hydrophilic R group (-0H), Alanine has hydrophobic R group ($-CH_3$)
ORE: Hence, serine is likely to face aqueous exterior and alanine in interior of globular protein.

4. The graph shows the amount of product formed by a standard concentration of enzyme and a standard concentration of substrate at a temperature of 15°C.



What would the rate be for an experimental temperature of 25°C

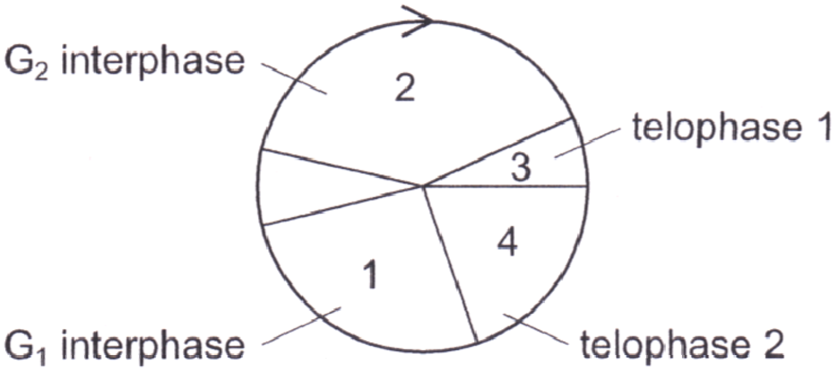
A 5 **B** 10 **C** 15 **D** 20

Ans: D

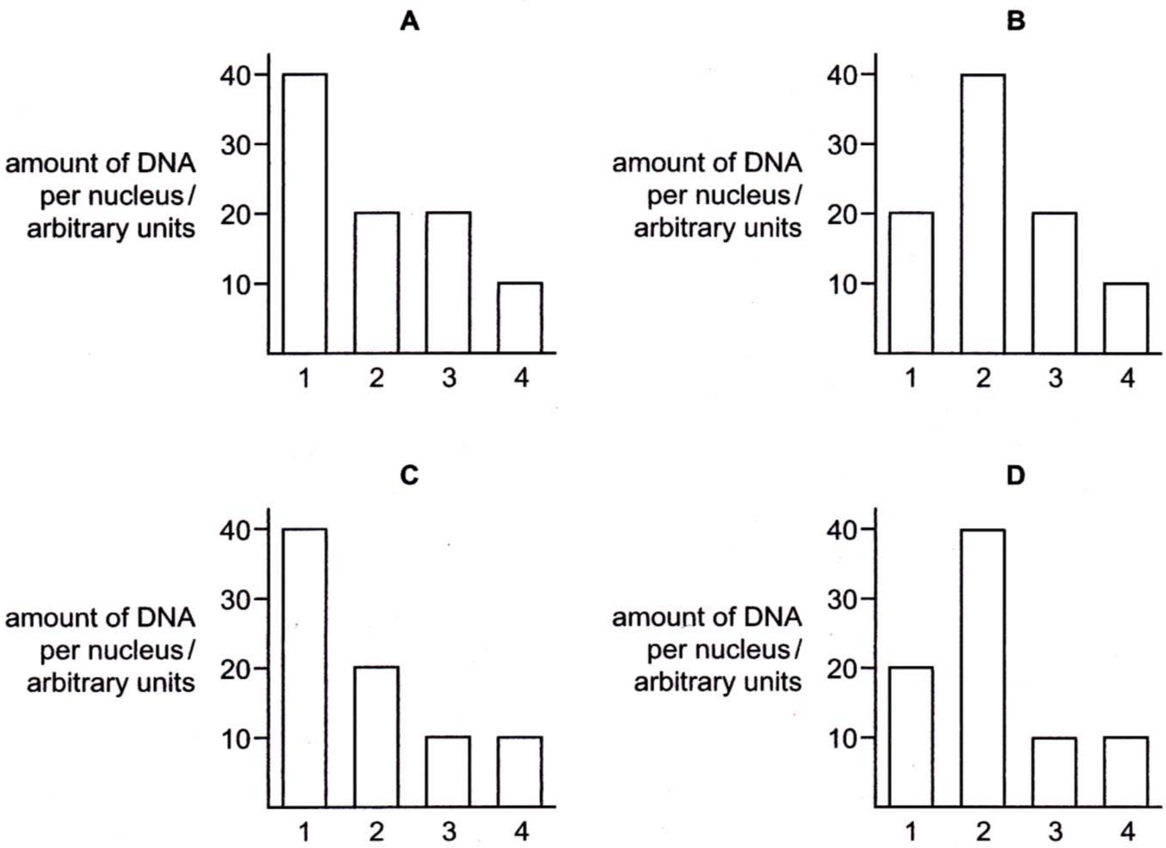
[L1] (*09/P1/ 3 modified)*

SC: Current temperature 15°C, with a rise to 25°C
ORE: there would be a doubling of the rate of reaction for every rise in 10°C (Q10)

5. The amount of DNA per nucleus in a cell during a meiotic cell cycle, as shown below, was measured.



Which bar graph correctly represents the variation in the DNA content?

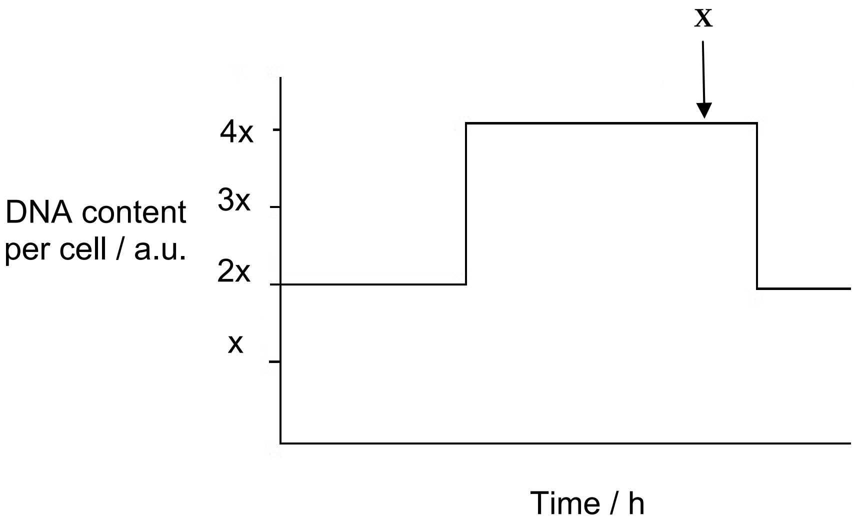


Ans: B

[L2] (*11SRJC/P1/6*)

SC: At G2 interphase, DNA has replicated hence double amount of DNA, at telophase 1, DNA reduced half due to meiosis I, DNA further halved due to meiosis II.
ORE: Option B selected.

6. The graph represents the changes in the quantity of DNA present in a cheek cell. The cell has 18 chromosomes at prophase.



Which of the following combinations is correct at stage **X**?

	Number of chromosomes	Number of centromeres	Process occurring at stage X
A	18	18	The daughter chromosomes start to loosen and uncoil.
B	36	36	The sister chromatids have separated and formed daughter chromosomes as they move towards the poles.
C	18	36	The daughter chromosomes start to loosen and uncoil.
D	36	18	The sister chromatids have separated and formed daughter chromosomes as they move towards the poles.

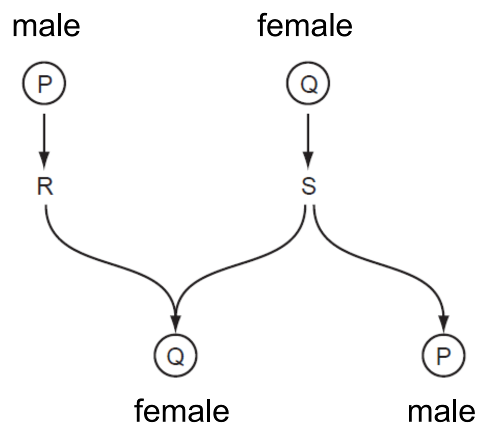
Ans: B

[L2] (*12HCI/P1/7*)

SC: No. of chromosomes will be double due to DNA replication during interphase. X is anaphase, hence no. of centromeres = no. of chromosome.
ORE: Option B selected.

7. Sex determination in some insects such as bees and wasps is not controlled by sex chromosomes.

Using the diagram, which row in the table shows how sex is determined in these insects?



	P (ploidy)	Q (ploidy)	R (process)	S (process)
A	Haploid	Haploid	mitosis	mitosis
B	Haploid	Diploid	mitosis	meiosis
C	Diploid	Haploid	meiosis	meiosis
D	Diploid	Diploid	meiosis	Mitosis

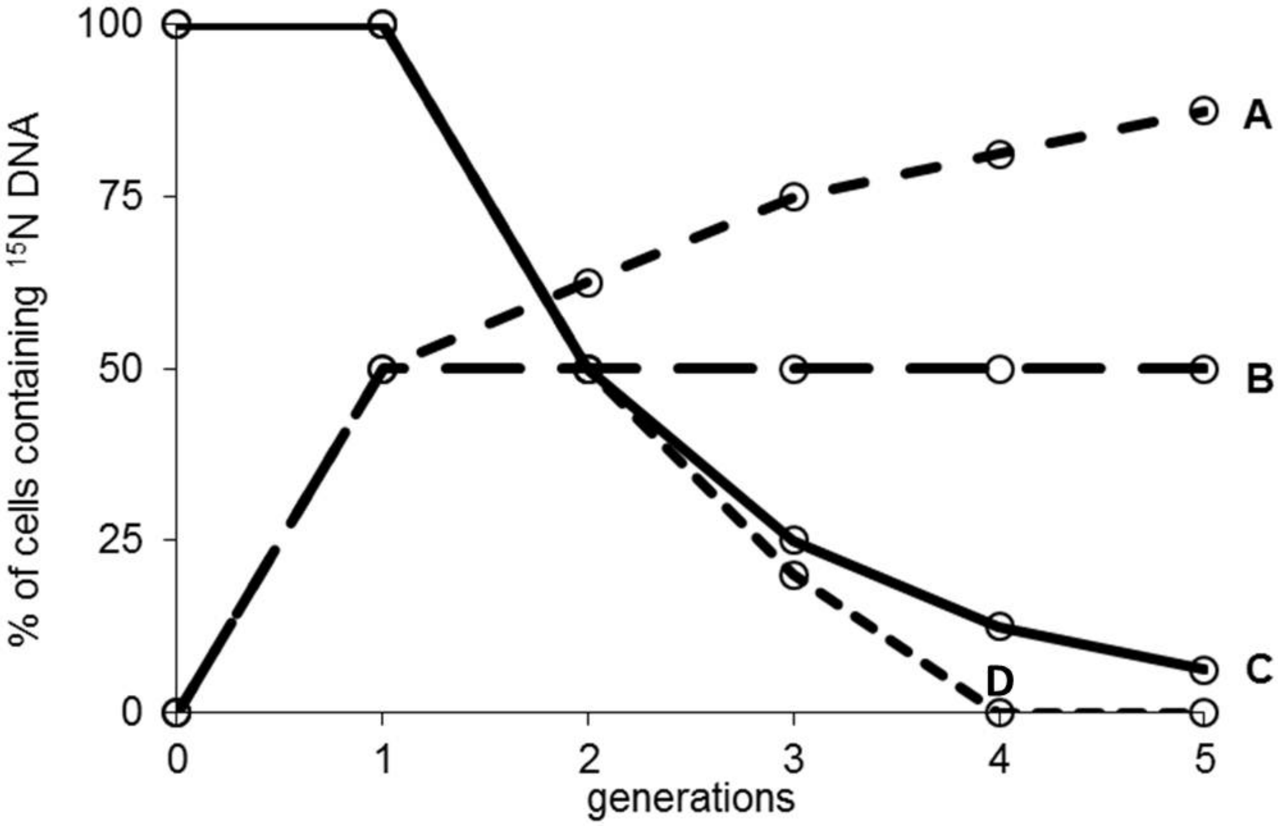
Ans: B

[L3] (*12RI/P1/7*)

SC: Q is likely to be diploid due to combination of product of R and S. P is likely to haploid.
ORE: Hence corresponding R should be mitosis and S meiosis.

8. The Bacteria were cultured in a medium containing heavy nitrogen (^{15}N) until all the DNA was labelled. These bacteria were then grown in a medium containing only normal nitrogen (^{14}N) for five generations. The percentage of cells containing ^{15}N in each generation was estimated.

Which curve provides evidence that DNA replication is semi-conservative?



Ans: C

[L2] (CJC/Prelim/2011/P1/Q7)

SC: Meselson & Stahl experiment. DNA replicates via semi-conservative replication.
ORE: % of **cell** containing ^{15}N DNA will reduce by half after the 1^{st} generation but will never reach 0 (option D). Therefore correct option is C.

9. Messenger RNA is isolated by passing homogenised tissue through a fractionating column. The column consists of short lengths of uracil nucleotides attached to a solid support medium. The mRNA that passes through the column cannot be translated as they break up into fragments. The mRNAs that attach to the column can be separated by application of sodium hydroxide.

Which statement is true of the mRNA in this experiment?

(i) Complementary base-pairing occurs between adenine and uracil bases in the column.

(ii) The mRNA that pass through the fractionating column probably lack poly A tails.

(iii) The mRNA that attach to the fractionating column can be translated.

(iv) The mRNA that attach to the fractionating column probably lack poly A tails.

A (i) and (iv)

B (i), (ii) and (iii)

C (ii) and (iv)

D All of the above.

Ans: B

[L3] (*11MJC/P1/17)*

SC: column has uracil attached hence will complementary base pair with adenine (trap mRNA with poly-A tails) → statements (i), (ii) and (iii) are true and statement (iv) false)
ORE: Option B selected.

10. Which of the following is not a requirement for RNA splicing?

A Spliceosome
B ATP
C RNA polymerase
D Splice site

Ans: C

[L1] (14*CJC/P1/10*)

SC: Option A, B and D unlikely (splicing require energy, spliceosome and splice site),
ORE: Hence Option C selected (RNA polymerase not involved directly for RNA splicing)

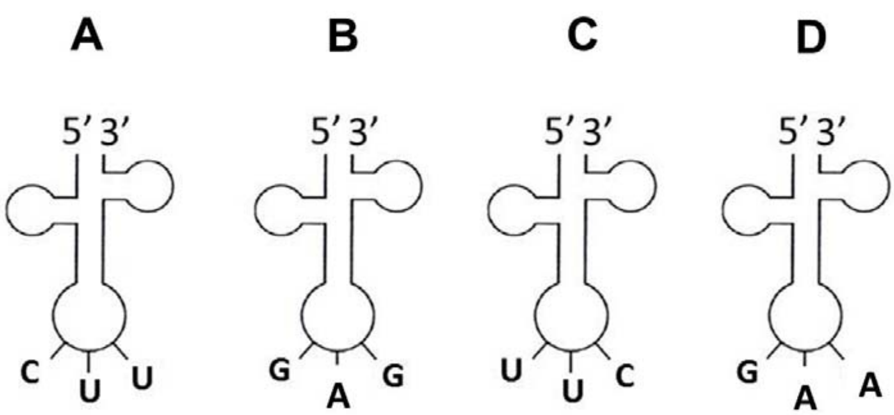
11. Part of the amino acid sequence of protein Z in normal and disease patients are shown below.

Protein Z in normal individual	Protein Z in diseased patient
…Lys-His-Pro-Asp-Thr…	…Lys-His-Pro-Glu-Thr…

mRNA codons for the amino acids shown in the sequence are as follows:

His: CAU, CAC	Asp: GAU, GAC	Lys: AAA, AAG
Thr: ACU, ACC, ACA, ACG	Glu: GAA, GAG	Pro: CCU, CCC, CCA, CCG

Which transfer RNA was utilised in forming the part of protein Z in diseased patients but not in normal individuals?



Ans: C

[L3] (modified from CJC/Prelim/2013/P1/Q7)

SC: Anticodon on tRNA binds to codon on mRNA.

ORE:

- Glu is present in diseased patient and not normal individual
- Codon for Glu is GAA & GAG
- Anticodon for Glu is therefore CUU & CUC
- Orientation should be correct so…..5'-UUC-3' is correct.

12. Two structural proteins, J and K, are each encoded by different alleles of the same gene. The amino acid sequences of the two structural proteins differ between positions 91 and 95 of the polypeptides.

The amino acid sequences of enzymes J and K, and the corresponding DNA sequence of structural protein J from position 90 to position 97 of the polypeptides, are shown in the table below.

	←N-terminal end			amino acid position				C-terminal end →
	90	91	92	93	94	95	96	97
DNA triplet codes for structural protein J	TTT	TCA	GGT	AGT	GAA	TTA	CGA	CGA
Amino acid sequence of structural protein J	lys	ser	pro	ser	leu	asn	ala	ala
Amino acid sequence of structural protein K	lys	val	his	his	leu	met	ala	ala

The actual mRNA codons for the amino acids in these positions for structural proteins J and K are shown in the table below.

Amino acid	lys	ser	pro	leu	asn	ala	val	his	met
mRNA codon (s)	AAA	AGU UCA	CCA	CUU UUA	AAU	GCU	GUC	CAU CAC	AUG

What could account for the difference in amino acid sequence of structural proteins J and K?

A Frame shift mutations in the DNA codes at position 91 and position 94.

B A deletion in the DNA code at position 91 and an insertion into the DNA code at position 96.

C A single frame shift by deletion in the DNA code at position 91.

D A change in the sequences of the second and third nucleotides at positions 91 and 92 of the DNA codes and frame shifts at positions 93 and 95.

Ans: B

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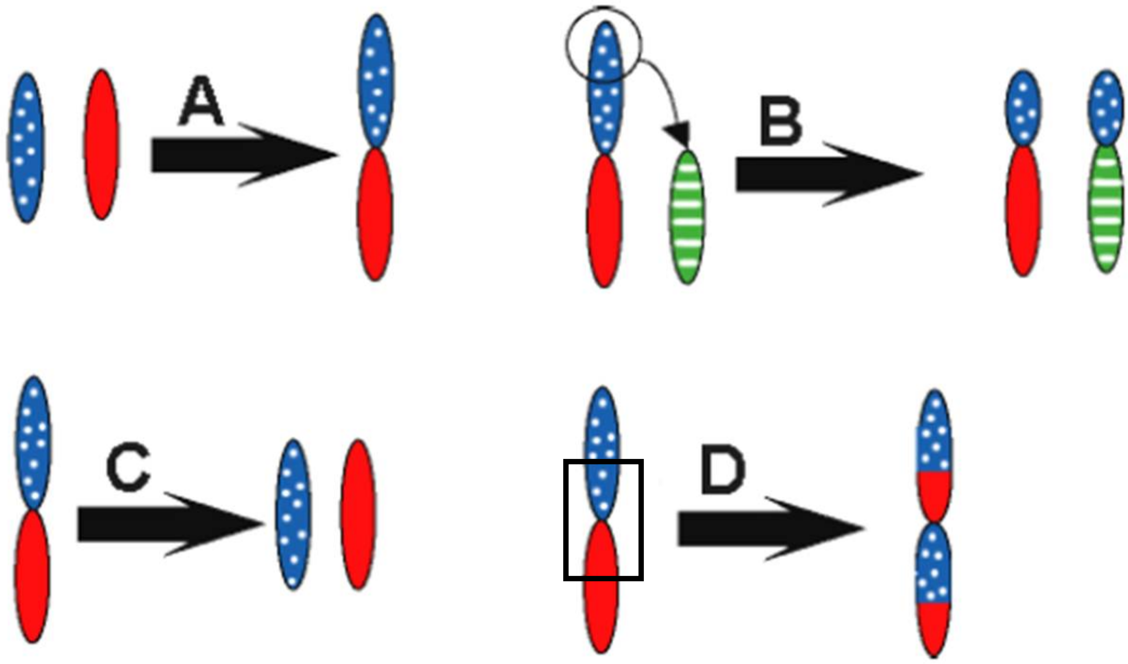
[L3] (adapted from A-level/H2/2013/P1/Q10)

SC: CW: What could account, QKW: differences in amino acid sequences, proteins J & K
ORE:

- Determine the codon sequence of structural proteins J & K.
- Compare the codon sequence of J & K.
- Identify the points where deletion and insertion mutations have occurred.

13. Which of the following illustrates chromosomal inversion?



Ans: D

[L2]

SC: Chromosomal inversion → involves a portion of the chromosome inverting.
ORE: Options A & B are translocation, Option C is deletion. Option D is correct.

14. In lentils, the seed coat pattern is determined by a gene with 3 alleles. The phenotypes are marbled, spotted and clear. 4 crosses were repeated many times. The crosses and the outcomes of these crosses are shown in the table below.

Cross	Parents	Offspring phenotype and ratio
1	Marbled X Marbled	3 marbled : 1 clear
2	Spotted X Clear	1 spotted : 1 clear
3	Marbled X Marbled	3 marbled : 1 spotted
4	Clear X Clear	All clear

From the data, it is possible to conclude that

A Spotted is recessive to clear.

B All the clear offspring are heterozygous.

C Two thirds of the marbled offspring in cross 3 are heterozygous.

D The marbled parents in cross 1 have the same genotype as the marbled parents in cross 3.

Ans: C

[L1] (HCI(H1 and H2)/1/16)

SC: Do a monohybrid cross.

ORE: All options are incorrect except statement C.

15. Red-green colour-blindness is a sex-linked, recessive trait. A colour-blind man marries a woman with normal vision, whose father is colour-blind. If they have a daughter, what is the probability that she will be colour-blind?

A 0

B ¼

C ½

D ¾

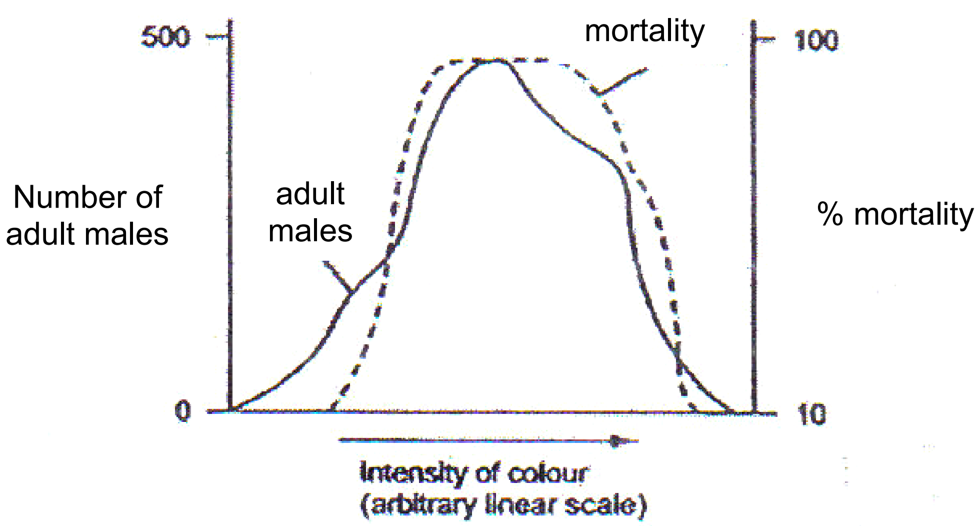
Ans: C

[L1] (ACJC(H1)/1/10)

SC: Do a sex-linked cross.

ORE: X^RX^r x X^rY will give you a 50% chance that the daughter is colour blind (X^rX^r).

16. The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity



Which conclusion can be made from this graph?

A Colour variation is environmentally induced.

B Colour variation is genetically determined.

C Extreme forms are favoured by natural selection.

D The species shows discontinuous variation with respect to colour.

Ans: C

[L3] (NYJC(H2)/1/31)

SC: Statements A and B non-conclusive from data given. Statement C is possible as extreme forms survive (low mortality). Statement D false as data shows continuous variation (normal distribution within population).
ORE: Option C selected.

17. Student **X**, looked after a plant whereas student **Y** looked after another plant of the same species. Each student followed the same instructions to set up apparatus to take cuttings from their plant and grow the cuttings next to the plant from which the cutting had been taken.

Fig. 17 shows the results after one week.

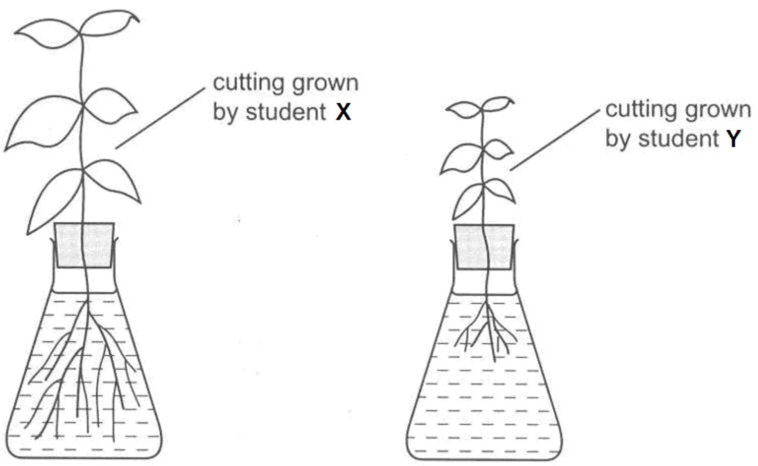


Fig. 17

Which factors could have caused the different appearance of the two cuttings?

1 phenotypic variation due to environment
2 genetically different cuttings
3 mutation due to environment
4 genotypic variation due to environment

A ***1*** and ***2*** **B** ***2*** and ***3*** **C** ***2*** and ***4*** **D** ***1***, ***3*** and ***4***

Ans: A

[L2] (*11RVHS/P1/24)*

SC: Qn context support statement 1 and 2. Statement 3 and 4 non-conclusive from data given.
ORE: Option A selected.

18. Which of the following processes do not require the utilisation of ATP.

A Active transport

B Endocytosis

C Facilitated diffusion

D Exocytosis

Ans: C

[L1] (14CJC/18))

SC: Bulk transport and transport across membranes
ORE: All except facilitated diffusion require ATP.

19. Which of the following processes could still occur in a chloroplast in the presence of an inhibitor that prevents H^+ from passing through ATP synthase complexes?

1 Sugar synthesis
2 Photolysis of water
3 Transfer of electrons down the electron transport chain
4 Oxidation of NADPH

A 1 and 2

B 1 and 4

C 2 and 3

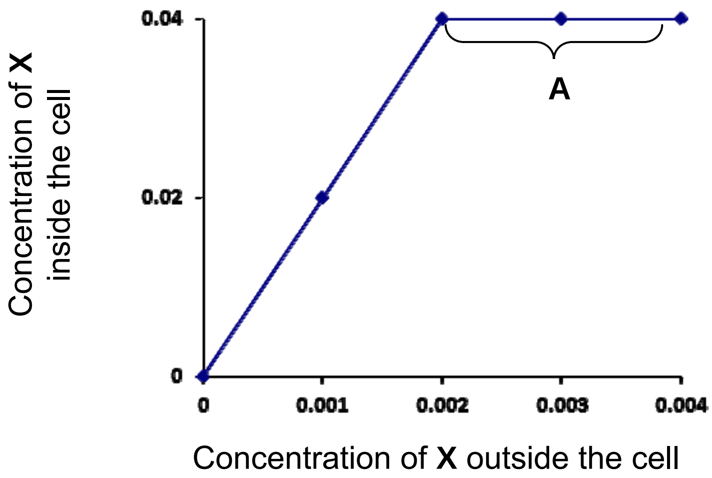
D 3 and 4

Ans: C

[L2] (ACJC(H1)/1/20)

SC: Light dependent and independent reactions
ORE: Sugar synthesis and oxidation of NADPH during Calvin cycle cannot occur in the absence of ATP, hence options 2 and 4 are wrong.

20. Substance **X** is actively transported into cells. Equal-sized samples of cells were placed in media containing different concentrations of **X** for an hour. The intracellular concentration of **X** was then measured. All other metabolic conditions were maintained at the optimum level. The graph below shows the results.



From the information given above, which one of the following would account for the region of the graph labelled A?

A A respiratory inhibitor had been introduced.

B The active transport carriers had been inactivated by a non-competitive inhibitor.

C All the active transport carriers had been operating at their maximum rate.

D As the internal concentration of **X** rose, more of the substance **X** was metabolised.

Ans: C

[L3] (*12AJC/P1/3*)

SC: Option A, B and D unlikely as it would see a decrease in concentration of X. Statement C possible
ORE: Option C selected.

21. Which of the following statements could NOT be used to describe a species?

A A group of organisms showing distinctly similar autosomes.

B A group of organisms showing analogous body structures.

C A group of organisms capable of mating to produce viable offspring.

D A group of organisms sharing the same ecological niche.

Ans: B

[L1] (TJC(H2)/1/33)

SC: Definition of species
ORE: A, B & D are correct. Only different species have analogous body structures.

22. Which effect of natural selection is likely to lead to speciation?

A Differences between populations are increased.

B Favourable genotypes are maintained in the population.

C Genetic diversity is reduced.

D Selection pressure on some alleles reduces reproductive success.

Ans: A

[L2] (*07P1/33*)

SC: effect Natural Selection Speciation
ORE: A] Correct- the greater the difference the more distinct populations are
B] allows survival and reproduction but may not lead to speciation
C] will reduce survival and may not lead to speciation
D] Reduction in reproductive success diminishes constancy of population and may not lead to speciation

23. A comparison was made between human, rabbit, mouse and chimpanzee in the

- DNA coding sequence of the β globin gene
- DNA sequence in the introns of the β globin gene
- Amino acid sequence of the β globin polypeptide

The data is shown in Table 1.

Table 1

	sequence similarity (%)		
Organisms being compared	**coding DNA**	**intron**	**amino acid sequence**
human β globin/chimpanzee β globin	100	98.4	100
human β globin/rabbit β globin	89.3	67	90.4
human β globin/mouse β globin	82.1	61	80.1

It is possible to conclude from this data that

A a human is more closely related to a mouse than to a rabbit.

B the variation between chimpanzees and human occurs in a region of the β globin gene which would code for amino acids.

C the variation in the intron sequence between human and mouse would account for some of the differences in the amino acid sequence.

D the comparison between human and chimpanzee indicates that human is more closely related to chimpanzee than to rabbit and mouse.

Ans: D

[L3] (TJC(H2)/1/33)

SC: Molecular homology analysis
ORE: D is the best option amongst the 4 as options A, B and C are conceptually wrong.

24. Which of the following enzyme is not used in DNA cloning?

A DNA ligase

B Restriction endonuclease

C Primase

D Reverse transcriptase

Ans: C

[L1] (14CJC/24)

SC: Enzymes involved in DNA cloning.
ORE: All the enzymes are involved in DNA cloning except Primase is used during DNA replication.

25. Which of the following primers would allow for the copying of the single-stranded DNA sequence, 5'-ATGCCTAGGTC -3' in the polymerase chain reaction (PCR)?

A 5' TACGG 3'

B 5' TCCAG 3'

C 5' GACCU 3'

D 5' GACCT 3'

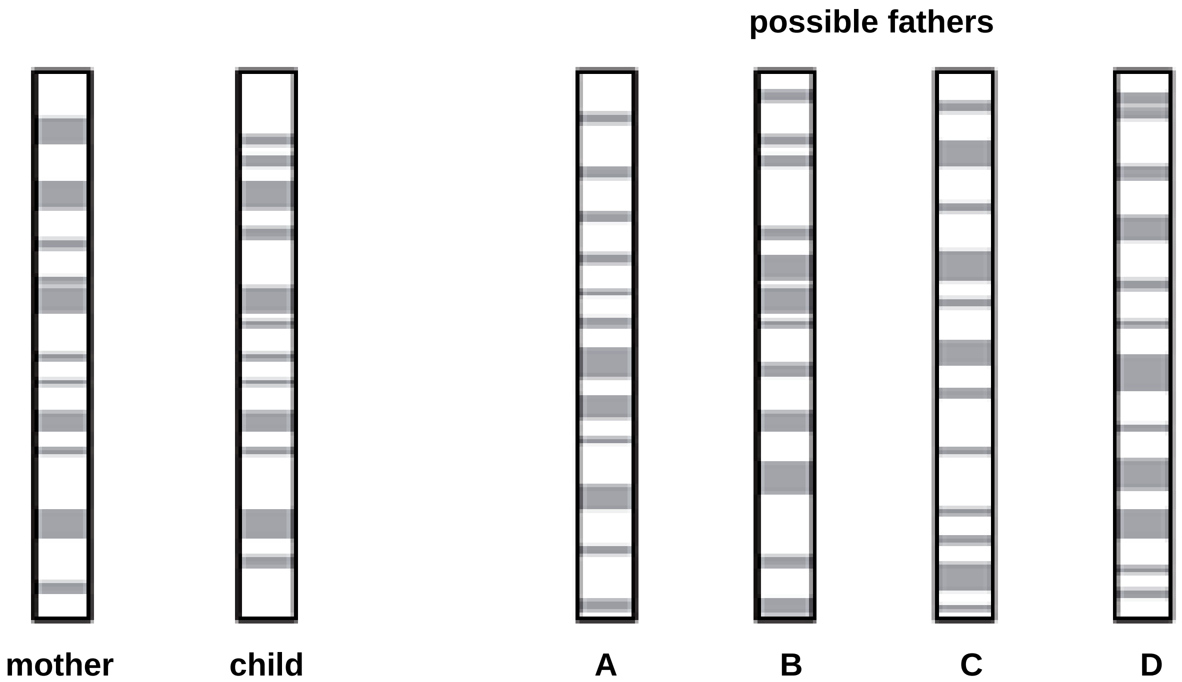
Ans: D

[L2] (*12RI/P1/34*)

SC: Primer should be annealing to 3' end of template sequence.
ORE: Option D selected based on antiparallel and complementary base-pairing.

26. Genetic profiling can be used to determine the paternity of a child. DNA from the mother and the child is cut into fragments, separated by electrophoresis and made visual using a stain. The diagram shows the genetic profiles of a mother and child, and four possible fathers.

Who is the father?



Ans: B

[L2] (*12RI/P1/35*)

SC: Child should have matching band profile with parent
ORE: B is likely the father due to presence of matching bands

27. What is not expected to be an outcome of the Human Genome Project?

- **A** Identification of all genes present in human beings
- **B** Identification of all alleles present in human beings
- **C** Map and sequence the genetic makeup of every human being
- **D** Map and sequence all the genetic material present in the chromosomes of human beings

Ans: C

[L1] (*N2007/1/28 (H1)*)

SC: Human genome project
ORE: The purpose of the HGP is not to map the genetic makeup of every single human being.

28. Which of the following is not a property of all stem cells?

- **A** Able to self-renew
- **B** Able to proliferate continually
- **C** Able to differentiate into specialised cells
- **D** Totipotency

Ans: D

[L1] (CJC/2014)

SC: Recall stem cells properties.
ORE:
- Option A, B & D: Correct properties for ALL stem cells.
- Option D: Not all stem cells are totipotent.

29. Which of the following statements could not explain why infection with *Agrobacterium tumefaciens* for genetic engineering of plant cells is preferred?

A *Agrobacterium tumefaciens* can efficiently transfer DNA into plant cells upon infection.

B *Agrobacterium tumefaciens* can harbour the Ti plasmid in which foreign genes can be inserted within it.

C Plant cell protoplasts may impede DNA entry via gene gun delivery.

D Plant cells have a cellulose cell wall which may impede DNA entry via transformation.

Ans: C

[L2] (CJC/2014/Prelim)

SC: QKW: infection with *Agrobacterium tumefaciens*, genetic engineering, plant cells, preferred
ORE:
- Option A: Infection of plant cells with bacteria is an efficient mode of gene delivery.
- Option B: *Agrobacterium tumefaciens* can be re-transformed with the Ti plasmid which contains foreign genes.
- Option C: Plant cell protoplast does not impede DNA entry via gene gun delivery.
- Option D: The plant cell wall does hamper transformation.

30. Normal tomato plants have a gene that codes for an enzyme that softens tomatoes as they ripen.

The DNA sequence for this gene is as follows:

3’ … T T A G C C T T A … 5’,

Genetically engineered tomatoes ripen and soften more slowly because of the insertion of a transgene which reduces the amount of the softening enzyme produced by hybridizing with mRNA encoding the enzyme.

Which of the following would be the mRNA sequence encoded by this transgene?

A 5’ – A U U C C G A U U – 3’

B 5’ – A A T C G G A A T – 3’

C 5’ – A A U C G G A A U – 3’

D 5’ – U U A G C C U U A – 3’

Ans: A

[L3] (NJC/2007/Prelim/P1/25)

SC: QKW: mRNA sequence encoded

ORE:
- Derive mRNA sequence from DNA sequence
- Find the antisense sequence of the mRNA sequence
- Check directionality (i.e. 5’ & 3’).

END OF PAPER